

Feasibility Study for a Smart Parking System

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COURSE CODE:CSA1006

COURSE: SOFTWARE ENGINEERING

ASSIGNMENT TOOL NO 1

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 **Smart Parking System**

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Smart Parking System

Feasibility Study for a Real-Time Smart Parking System using IoT Sensors and Mobile Booking Application

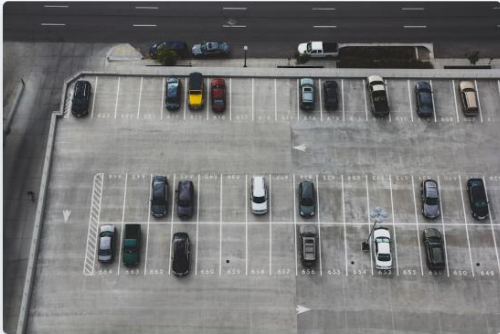
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Project Overview



Introduction

Project Overview



Introduction

The Smart Parking System is designed to help drivers identify available parking spaces using IoT sensors and a mobile application. It reduces traffic congestion, improves parking efficiency, and provides real-time parking information.

Objectives

- Reduce parking search time
- Provide real-time parking availability
- Improve customer satisfaction
- Increase parking efficiency
- Support online slot booking

Technical Feasibility

Hardware

- IoT Parking Sensors
- CCTV Cameras
- GPS Module
- Parking Display Board
- RFID Reader

Software

- Mobile Application
- Cloud Database
- Admin Dashboard
- Notification Service
- Google Maps API

Network

- Wi-Fi / Internet
- Cloud Hosting
- Real-Time Communication
- GPS Connectivity
- Secure Data Transfer

Technical Feasibility Summary: The Smart Parking System is technically feasible because all required hardware, software and networking technologies are already available. IoT sensors, cloud computing and mobile applications make the project easy to implement and maintain.

Operational Feasibility

Operational Feasibility

Benefits

- Easy to use mobile application
- Real-time parking updates
- Less manual work
- Faster parking process
- Improved customer satisfaction

Challenges

- Internet dependency
- Sensor maintenance
- Initial staff training
- Power backup requirement
- Data security

Operational Feasibility Report: The proposed Smart Parking System is operationally feasible because drivers and parking operators can easily use the application with minimum training. The system improves efficiency while reducing manual errors and waiting time.

Financial Feasibility

Item	Estimated Cost
IoT Sensors	₹40,000
Mobile App Development	₹60,000
Cloud Server	₹15,000
Database	₹5,000
Testing	₹10,000
Total	₹1,30,000

Design Thinking Activity

Empathy Interview

To understand user needs, an empathy interview was conducted with regular drivers and parking operators. Participants were asked about their parking experiences and common challenges.

Sample Questions

- How often do you struggle to find parking?
- How much time do you spend searching for a parking space?
- Would you use a mobile app to reserve parking?
- What features would improve your parking experience?

Findings

- Users want real-time parking availability.
- Drivers prefer advance slot booking.
- Navigation to the selected parking space is important.
- Online payment and booking confirmation are useful features.

Conclusion: The empathy interview helped identify user requirements and guided the design of a user-friendly Smart Parking System.

Cost-Benefit Analysis

Estimated Costs

Expected Benefits

Cost-Benefit Analysis

Estimated Costs

- IoT Sensors - ₹40,000
- Mobile App Development - ₹60,000
- Cloud Server - ₹15,000
- Database - ₹5,000
- Testing & Maintenance - ₹10,000

Expected Benefits

- Reduced traffic congestion
- Faster parking allocation
- Increased parking revenue
- Better customer satisfaction
- Efficient parking management

Cost-Benefit Estimation: Although the project requires an initial investment of approximately ₹1,30,000, the long-term operational benefits, reduced manpower, and improved efficiency make it a profitable solution.

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Name

Email

Message

Cost-Benefit Estimation: Although the project requires an initial investment of approximately ₹1,30,000, the long-term operational benefits, reduced manpower, and improved efficiency make it a profitable solution.

Contact Us

Name
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Message
estimatees cost is ok i will look on it ...can i make it use of the cloud sever->

Submit

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• Testing & Maintenance - ₹10,000

• Efficient parking management

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Thank You, Bavani Sri!

Your feedback has been submitted successfully.

OK

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